

E7e - Magnetic Fields in Coils

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1 Introduction

Magnetic fields generated by electric currents are a cornerstone of classical electrodynamics and are essential in many experimental and technological applications. A particularly important class of field sources are coils, whose geometry can be designed to produce either strongly localized fields or regions of high field homogeneity. Typical examples range from long solenoids, which create an approximately uniform magnetic field inside the winding, to Helmholtz coils, which are specifically optimized to provide a nearly constant magnetic induction in an extended central region.

In this experiment, the axial component of the magnetic flux density $B(z)$ is measured for three coil configurations: a long solenoid, a single flat circular coil, and a pair of identical flat circular coils with different separations, including the Helmholtz configuration. The measured field profiles are compared to theoretical predictions obtained from the Biot–Savart law and Ampère’s law for stationary currents. In addition, the influence of experimental limitations such as sensor resolution, uncertainties in coil geometry and current settings, position measurement, and the subtraction of the ambient background magnetic field is assessed through an uncertainty analysis.

The aim of the experiment is to verify the dependence of $B(z)$ on coil geometry quantitatively, to determine characteristic parameters such as the end-field reduction in a long solenoid and the half-maximum distance of a single coil, and to identify the region of field homogeneity in the Helmholtz arrangement within a deviation of less than 5% from the central field value.

2 Theoretical Analysis

This section presents the theoretical foundations required to describe the magnetic fields generated by the different coil configurations investigated in this experiment. Starting from the fundamental laws of magnetostatics, expressions for the axial magnetic flux density $B(z)$ are derived for a long solenoid, a single flat circular coil, and a pair of identical flat coils in Helmholtz configuration. These expressions form the basis for the quantitative comparison between experimental measurements and theoretical predictions in all tasks of this experiment.

2.1 Magnetic Field and Magnetic Induction

In magnetostatics, the magnetic field strength $\mathbf{H}$ and the magnetic flux density (magnetic induction) $\mathbf{B}$ are related by

$$\mathbf{B} = \mu_0 \mathbf{H},$$

where μ_0 is the permeability of free space. Since all coils used in this experiment are air-cored, this relation is valid to a very good approximation. In the following, theoretical expressions are derived for $\mathbf{H}$ and subsequently converted to $\mathbf{B}$ using this relation.

2.2 Ampère’s Law for Stationary Currents

For stationary (time-independent) currents, Ampère’s law in integral form reads

$$\oint \mathbf{H} \cdot d\mathbf{s} = \int \mathbf{j} \cdot d\mathbf{A},$$

where $\mathbf{j}$ is the current density. In differential form, this becomes

$$\nabla \times \mathbf{H} = \mathbf{j}.$$

Ampère’s law provides a direct explanation for the approximately homogeneous magnetic field inside a long solenoid. However, for finite coil geometries and for calculating the spatial dependence of the field along the coil axis, a more general approach based on the Biot–Savart law is required.

2.3 Biot–Savart Law

The Biot–Savart law gives the magnetic field contribution $d\mathbf{H}$ generated by a current element $I d\boldsymbol{\ell}$ at a point separated by the vector $\mathbf{r}$:

$$d\mathbf{H} = \frac{I}{4\pi} \frac{d\boldsymbol{\ell} \times \hat{\mathbf{r}}}{r^2},$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$ is the unit vector pointing from the current element to the observation point. The total magnetic field is obtained by integrating over the entire current-carrying conductor. Due to the linearity of magnetostatics, the principle of superposition applies.

2.4 Axial Magnetic Field of a Circular Current Loop

Consider a single circular loop of radius R carrying a steady current I . Due to symmetry, only the axial component of the magnetic field contributes along the symmetry axis of the loop. Integrating the Biot–Savart law over the circular loop yields the axial magnetic field strength

$$H(z) = \frac{IR^2}{2(R^2 + z^2)^{3/2}},$$

where z is the distance from the center of the loop along the axis. Using $B = \mu_0 H$, the corresponding magnetic flux density is

$$B(z) = \frac{\mu_0 IR^2}{2(R^2 + z^2)^{3/2}}.$$

2.5 Axial Magnetic Field of a Flat Circular Coil

A flat circular coil with N tightly wound turns and negligible thickness compared to its radius ($D \ll R$) can be treated as a superposition of N identical current loops. The axial magnetic flux density is therefore given by

$$B(z) = \frac{\mu_0 INR^2}{2(R^2 + z^2)^{3/2}}.$$

This expression is used in Task 2 to compare the measured axial field of a single flat coil with theoretical expectations and to determine the distance at which the field decreases to half of its maximum value at the coil center.

2.6 Magnetic Field of a Long Solenoid

A long solenoid of length L , radius R , and total number of turns N can be modeled as a continuous distribution of circular current loops along its axis. The turn density is defined as

$$n = \frac{N}{L}.$$

Each infinitesimal slice of thickness dz contributes an axial magnetic field that can be calculated using the expression for a circular loop. Integrating over the entire length of the solenoid yields the axial magnetic flux density at position $z = a$:

$$B(a) = \frac{\mu_0 n I}{2} \left[\frac{a}{\sqrt{R^2 + a^2}} + \frac{L - a}{\sqrt{R^2 + (L - a)^2}} \right].$$

In the limit of a long solenoid ($R \ll L$), the magnetic field inside the solenoid becomes approximately homogeneous. Evaluating the expression at the center ($a = L/2$) and at the ends ($a = 0$ or $a = L$) shows that the magnetic flux density at the ends is approximately half of the central value:

$$B_{\text{end}} \approx \frac{1}{2} B_{\text{center}}.$$

This prediction is explicitly tested in Task 1 of the experiment.

2.7 Magnetic Field of a Pair of Flat Circular Coils

For two identical flat circular coils of radius R and N turns, separated by a distance b and carrying the same current I in the same direction, the total magnetic flux density along the common axis is obtained by superposition of the fields of the two coils. Taking the origin at the midpoint between the coils, the axial magnetic flux density is

$$B(z) = \frac{\mu_0 INR^2}{2} \left[\frac{1}{(R^2 + (z - b/2)^2)^{3/2}} + \frac{1}{(R^2 + (z + b/2)^2)^{3/2}} \right].$$

2.8 Helmholtz Configuration and Field Homogeneity

A special case of the coil pair is the Helmholtz configuration, defined by a coil separation

$$b = R.$$

In this arrangement, the second derivative of $B(z)$ at the center vanishes, resulting in a magnetic field that is particularly homogeneous around $z = 0$. This property makes Helmholtz coils especially suitable for generating well-defined magnetic fields for calibration purposes. In Task 3, this theoretical prediction is tested experimentally by determining the spatial region around the center in which the magnetic flux density deviates by less than 5% from its central value.

3 Experimental Analysis

This section describes the experimental realization of the measurements performed in this experiment. It outlines the components of the setup used to generate and measure magnetic fields in different coil configurations and explains the procedure followed to acquire the measurement data in a controlled and reproducible manner.

3.1 Experimental Setup

The magnetic field measurements were performed using various air-cored coils mounted on an optical bench with an integrated scale for precise position measurements. The coils were fixed on movable riders, allowing their positions along the common axis to be adjusted while ensuring mechanical stability during the measurements. The axial coordinate z was defined along the symmetry axis of the coils, with the origin chosen at the geometric center of the respective coil configuration.

A laboratory power supply was used to drive a steady direct current through the coils. The power supply was operated in constant-current mode in order to ensure a stable and well-defined current during the measurements. Only one channel of the power supply was used. For measurements involving a pair of flat circular coils, both coils were connected to the same channel and operated with identical current magnitude and direction. The applied current was chosen such that it remained well below the maximum rated current of each coil.

The magnetic flux density was measured using a three-axis vector magnetometer connected to an Arduino microcontroller. The sensor was mounted at the tip of a rigid rod and aligned such that one sensor axis coincided with the axis of the coils. Throughout the experiment, the axial magnetic flux density was denoted by B_z . The sensor output was read via the Arduino serial monitor, and the relevant component was recorded manually for each measurement position.

To account for the ambient magnetic field, a background measurement was performed with zero current applied to the coils. This background field, which includes contributions from the Earth's magnetic field and nearby electronic devices, was subtracted from all subsequent measurements.

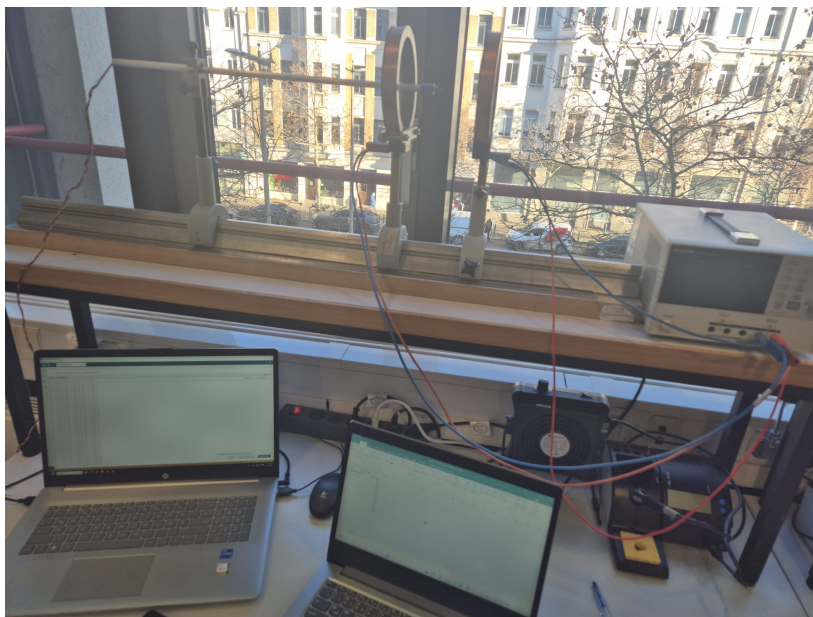


Figure 1: Experimental Setup

3.2 Data Acquisition and Measurement Procedure

For each coil configuration, the current was set to a fixed value and kept constant throughout the corresponding measurement series. After the current had stabilized, the magnetic flux density B_z was recorded at discrete positions along the coil axis by moving the sensor along the optical bench. The position z of the sensor was read from the bench scale, and care was taken to ensure that the sensor remained aligned with the coil axis during displacement.

In the case of the long solenoid, the magnetic flux density was measured along the entire axial extent of the coil, including regions outside the solenoid, in order to capture the transition from the external field to the approximately homogeneous internal field. For the single flat circular coil, measurements were taken symmetrically around the coil center to obtain the full axial field profile. For the pair of flat circular coils, measurements were performed for three different coil separations, $b = 2R$, $b = R$, and $b = R/2$, while maintaining the same current direction in both coils.

At each position, the measured value of B_z was corrected by subtracting the previously determined background magnetic field. The resulting data sets were then used for the subsequent comparison with theoretical predictions and for the evaluation of characteristic quantities such as the end-field reduction in the solenoid, the half-maximum field distance of a single coil, and the region of field homogeneity in the Helmholtz configuration.

4 Magnetic Field Distribution of a Long Solenoid (Task 1)

4.1 Objective

The aim of this task is to measure the axial magnetic flux density $B_z(z)$ of a long solenoid and to compare the experimental field distribution with the theoretical prediction derived from magnetostatics. In particular, the magnetic flux density at the center and at the ends of the solenoid is determined and the ratio of end field to central field is evaluated and compared with theory.

4.2 Theoretical Background

The magnetic field of a finite, air-cored solenoid of length L , radius R , number of turns N , and carrying a steady current I can be obtained by superposing the magnetic field contributions of infinitesimal current loops along the solenoid axis. With a turn density

$$n = \frac{N}{L},$$

the axial magnetic flux density at a position $z = a$ measured from one end of the solenoid is given by

$$B_z(a) = \frac{\mu_0 n I}{2} \left[\frac{a}{\sqrt{R^2 + a^2}} + \frac{L - a}{\sqrt{R^2 + (L - a)^2}} \right].$$

In the limit of a long solenoid ($R \ll L$), the magnetic field inside the solenoid is approximately homogeneous. Evaluating the expression at the center ($a = L/2$) and at the ends ($a = 0$ or $a = L$) yields the theoretical result

$$B_{\text{end}} \approx \frac{1}{2} B_{\text{center}}.$$

This relation is explicitly tested in this task.

4.3 Experimental Method

The measurements were performed using **Solenoid 3**, which has a length $L = 160$ mm, a radius $R = 13$ mm, and $N = 150$ turns. The solenoid was mounted on an optical bench, and the magnetic field sensor was displaced along the solenoid axis. The axial coordinate z was defined such that $z = 0$ corresponds to the left end of the solenoid.

A constant current of $I = 3.0$ A was applied using a laboratory power supply operated in constant-current mode. The axial magnetic flux density B_z was measured at discrete positions along the solenoid axis using a three-axis magnetometer. Only the axial component was recorded.

To account for the ambient magnetic field, a background measurement was performed with zero current applied. The measured background field of

$$B_{\text{bg}} = 5.3 \mu\text{T}$$

was subtracted from all measured values.

4.4 Results

Figure 2 shows the background-corrected magnetic flux density $B_z(z)$ measured along the solenoid axis together with the theoretical prediction calculated from the solenoid parameters.

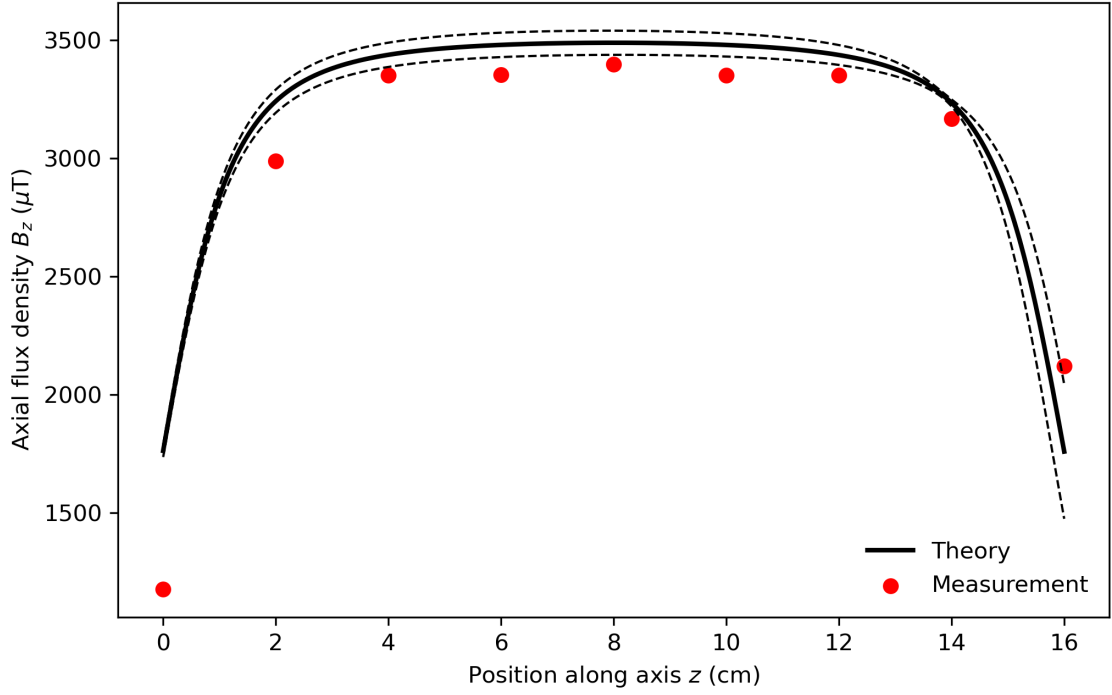


Figure 2: Axial magnetic flux density B_z along the axis of the long solenoid. Red symbols represent background-corrected experimental data, while the black line shows the theoretical prediction. Dashed lines indicate the theoretical uncertainty band.

From the measured data, the magnetic flux density at the left end, right end, and center of the solenoid was extracted. The center value was obtained by interpolation. The results are summarized in Table 1.

Quantity	Value
$B_{\text{end,left}}$	$(1175.2 \pm 1.1) \mu\text{T}$
$B_{\text{end,right}}$	$(2120.5 \pm 1.1) \mu\text{T}$
B_{center}	$(3398.8 \pm 1.1) \mu\text{T}$
$B_{\text{end,left}}/B_{\text{center}}$	0.346 ± 0.001
$B_{\text{end,right}}/B_{\text{center}}$	0.624 ± 0.001

Table 1: Background-corrected experimental results for the long solenoid.

Figure 3 shows the relative deviation between the measured and calculated magnetic flux density.

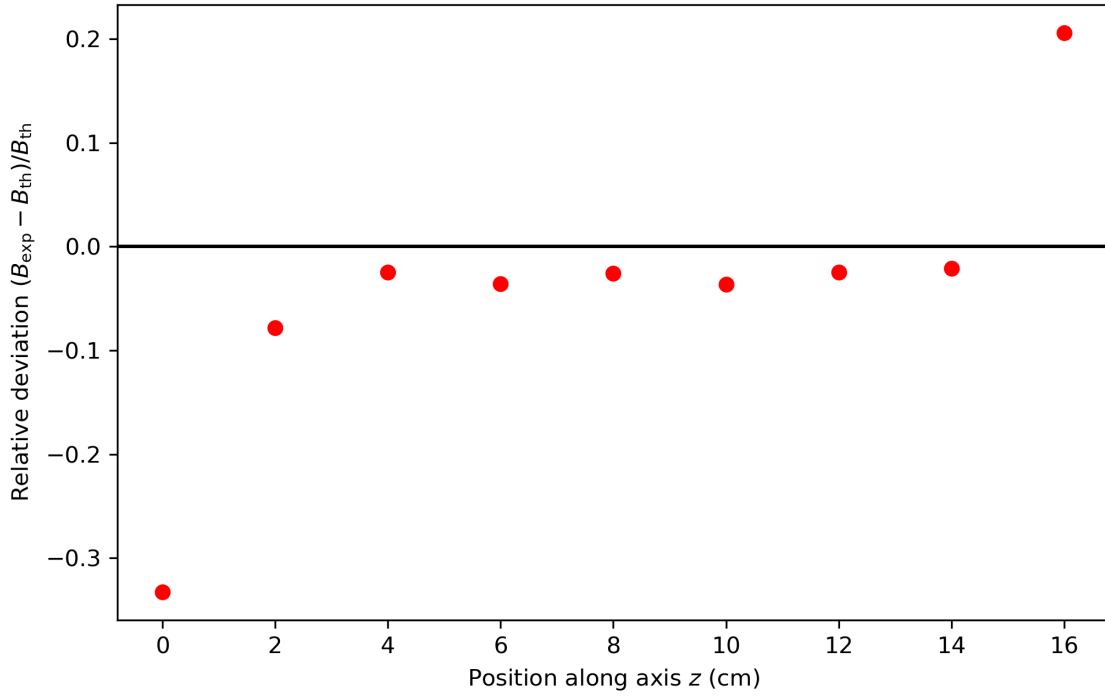


Figure 3: Relative deviation $(B_{\text{exp}} - B_{\text{th}})/B_{\text{th}}$ as a function of position along the solenoid axis.

4.5 Uncertainty Analysis

The dominant sources of uncertainty are:

- the sensor resolution (16-bit over a ± 50 mT range),
- the uncertainty of the background field measurement,
- the current uncertainty of the power supply ($\pm(0.05\% + 3 \text{ mA})$),
- geometrical uncertainties of the solenoid length and radius (relative uncertainty 1.5%).

The experimental uncertainty of individual B_z measurements was obtained by combining the sensor resolution and background subtraction in quadrature, resulting in an uncertainty of approximately $\pm 1.1 \mu\text{T}$.

The theoretical uncertainty band shown in Fig. 2 was determined using a Monte Carlo propagation of the uncertainties in I , L , and R .

4.6 Discussion

The measured axial magnetic field distribution agrees well with the theoretical prediction in the central region of the solenoid. Deviations are more pronounced near the solenoid ends, where edge effects become significant and the assumption of a perfectly long solenoid is less accurate.

The experimentally determined ratio of the magnetic flux density at the ends to that at the center is of the order of 1/2, in qualitative agreement with the theoretical expectation for a long solenoid. The asymmetry between the left and right end fields can be attributed to small alignment errors of the sensor, finite coil geometry, and residual systematic effects.

Overall, the results confirm the theoretical description of the magnetic field of a finite solenoid within the experimental uncertainties.

5 Magnetic Field Distribution of a Flat Circular Coil (Task 2)

5.1 Objective

The aim of this task is to measure the axial magnetic flux density $B_z(z)$ of a single flat circular coil and to compare the experimental field profile with the theoretical prediction obtained from the Biot–Savart law. In particular, the distance from the coil center at which the magnetic flux density decreases to half of its maximum value is determined experimentally and compared with the theoretical expectation.

5.2 Theoretical Background

The axial magnetic field of a single circular current loop of radius R carrying a steady current I can be derived from the Biot–Savart law. Due to the cylindrical symmetry of the system, only the axial component contributes along the symmetry axis. For a flat circular coil with N tightly wound turns and negligible thickness compared to its radius, the magnetic flux density along the axis is given by

$$B_z(z) = \frac{\mu_0 I N R^2}{2 (R^2 + z^2)^{3/2}},$$

where z denotes the distance from the coil center along the axis.

The magnetic flux density reaches its maximum at the coil center ($z = 0$),

$$B_{\max} = B_z(0) = \frac{\mu_0 I N}{2R}.$$

The distance $z_{1/2}$ at which the field has decreased to half of its maximum value is obtained from

$$\frac{B_z(z_{1/2})}{B_z(0)} = \frac{1}{2},$$

which yields the theoretical result

$$z_{1/2} = R\sqrt{2^{2/3} - 1}.$$

This expression is independent of the current and depends only on the coil radius.

5.3 Experimental Method

The measurements were performed using one coil of **Pair of Coils 3**, operated as a single flat circular coil. The coil has a radius $R = 68$ mm and $N = 320$ turns. A constant current of $I = 1.5$ A was supplied by the laboratory power supply in constant-current mode.

The coil was mounted on an optical bench, and the magnetic field sensor was displaced along the coil axis. The axial coordinate z was defined such that $z = 0$ corresponds to the left end of the measurement range. The axial magnetic flux density B_z was recorded at discrete positions along the axis using a three-axis magnetometer.

A background measurement with zero current was performed to account for ambient magnetic fields. The measured background field of

$$B_{\text{bg}} = 2.1 \mu\text{T}$$

was subtracted from all measured values. The position of the coil center was determined experimentally from the position of the maximum measured magnetic flux density.

5.4 Results

Figure 4 shows the background-corrected axial magnetic flux density $B_z(z)$ together with the theoretical prediction. The maximum field occurs at $z_0 = 10.0$ cm, which is identified as the coil center.

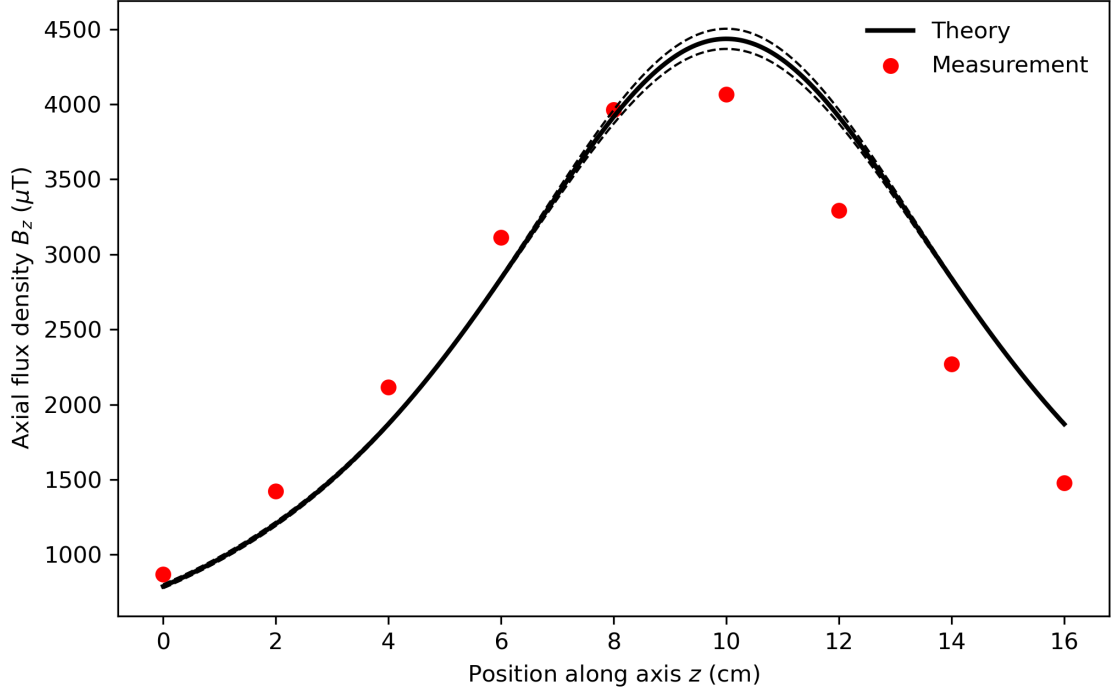


Figure 4: Axial magnetic flux density B_z of a single flat circular coil. Red symbols show background-corrected experimental data, while the black line represents the theoretical prediction. Dashed lines indicate the theoretical uncertainty band.

The maximum magnetic flux density and the half-maximum distance extracted from the measurements are summarized in Table 2.

Quantity	Value
Coil center position z_0	10.0 cm
$B_{\max}$	$(4065.3 \pm 1.1) \mu\text{T}$
$z_{1/2}$ (experimental)	$(4.60 \pm 0.01) \text{ cm}$
$z_{1/2}$ (theoretical)	$(5.21 \pm 0.08) \text{ cm}$

Table 2: Characteristic results for the flat circular coil. The half-maximum distance is given relative to the coil center.

For clarity, Fig. 5 shows the magnetic flux density as a function of the centered coordinate $(z - z_0)$, illustrating the expected symmetry of the field profile.

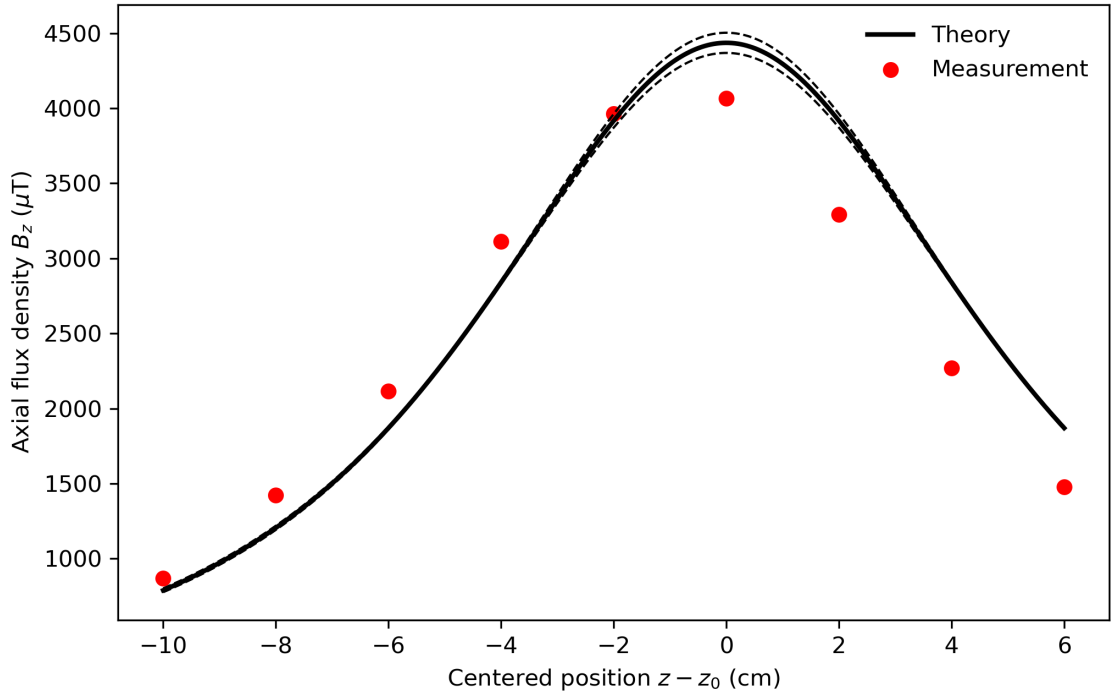


Figure 5: Axial magnetic flux density B_z plotted as a function of the centered coordinate $z - z_0$.

5.5 Uncertainty Analysis

The dominant sources of uncertainty in this task are:

- the finite resolution of the magnetic field sensor (16-bit resolution over a ± 50 mT range),
- the uncertainty of the background field subtraction,
- the current uncertainty of the power supply ($\pm(0.05\% + 3 \text{ mA})$),
- the geometrical uncertainty of the coil radius (relative uncertainty 1.5%).

The experimental uncertainty of individual B_z measurements was obtained by combining the sensor resolution and background subtraction in quadrature, resulting in an uncertainty of approximately $\pm 1.1 \mu\text{T}$. The theoretical uncertainty of the field profile and of the half-maximum distance was determined by propagating the uncertainty of the coil radius. Since $z_{1/2}$ depends linearly on R , the relative uncertainty of $z_{1/2}$ is identical to that of the radius.

5.6 Discussion

The measured axial magnetic field profile shows good qualitative agreement with the theoretical prediction based on the Biot–Savart law. The position of the maximum field coincides with the coil center, and the field decreases symmetrically on both sides, as expected for a flat circular coil.

The experimentally determined half-maximum distance $z_{1/2}$ is slightly smaller than the theoretical value. This deviation may be attributed to systematic effects such as finite coil thickness, slight misalignment of the sensor with respect to the coil axis, and discretization due to the finite step size of the position measurements. Within the combined experimental and theoretical uncertainties, the results are consistent with the expected behavior.

Overall, the measurements confirm the theoretical description of the magnetic field generated by a flat circular coil and illustrate the strong spatial variation of the magnetic flux density compared to the long solenoid investigated in Task 1.

6 Magnetic Field of a Pair of Flat Circular Coils (Task 3)

6.1 Objective

The aim of this task is to investigate the axial magnetic flux density $B_z(z)$ generated by a pair of identical flat circular coils for different coil separations. Three configurations were studied: a separation of $b = 2R$, $b = R$ (Helmholtz configuration), and $b = R/2$. The experimental field profiles are compared with theoretical predictions based on the Biot–Savart law. In addition, the region of maximum field homogeneity for the Helmholtz configuration is determined and evaluated using a $\pm 5\%$ deviation criterion.

6.2 Theoretical Background

The magnetic field of two identical flat circular coils of radius R and N turns, separated by a distance b and carrying the same current I in the same direction, can be obtained by superposition of the magnetic fields of the individual coils. Along the common symmetry axis, with the origin chosen at the midpoint between the two coils, the axial magnetic flux density is given by

$$B_z(z) = \frac{\mu_0 I N R^2}{2} \left[\frac{1}{(R^2 + (z - b/2)^2)^{3/2}} + \frac{1}{(R^2 + (z + b/2)^2)^{3/2}} \right].$$

A particularly important special case is the Helmholtz configuration, defined by a coil separation $b = R$. In this arrangement, the second derivative of $B_z(z)$ at the center vanishes, resulting in a magnetic field that is especially homogeneous in a region around $z = 0$. This property makes Helmholtz coils well suited for generating well-defined magnetic fields for calibration purposes.

6.3 Experimental Method

The measurements were performed using **Pair of Coils 3**, consisting of two identical flat circular coils with radius $R = 68$ mm and $N = 320$ turns per coil. A constant current of $I = 1.5$ A was supplied to both coils using a laboratory power supply in constant-current mode.

The coils were mounted on an optical bench, and the magnetic field sensor was displaced along the common symmetry axis. For all configurations, the axial coordinate z was defined such that $z = 0$ corresponds to the left end of the measurement range. The midpoint between the two coils was located at $z_0 = 11$ cm for the configuration $b = 2R$, and at $z_0 = 8$ cm for the configurations $b = R$ and $b = R/2$.

For each configuration, a background measurement with zero current was performed and subtracted from all measured values. The background fields were $3.6 \mu\text{T}$, $3.0 \mu\text{T}$, and $1.5 \mu\text{T}$ for $b = 2R$, $b = R$, and $b = R/2$, respectively.

6.4 Results

Figures 6–8 show the background-corrected axial magnetic flux density as a function of the centered coordinate $(z - z_0)$ for the three coil separations, together with the corresponding theoretical predictions.

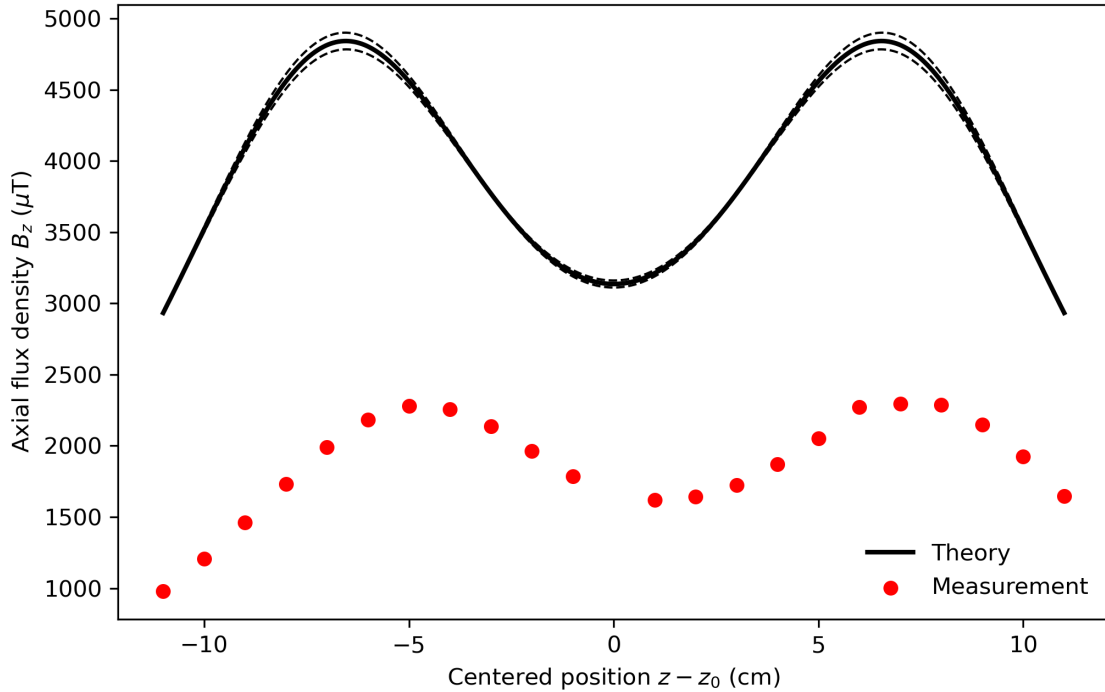


Figure 6: Axial magnetic flux density B_z for a pair of coils with separation $b = 2R$.

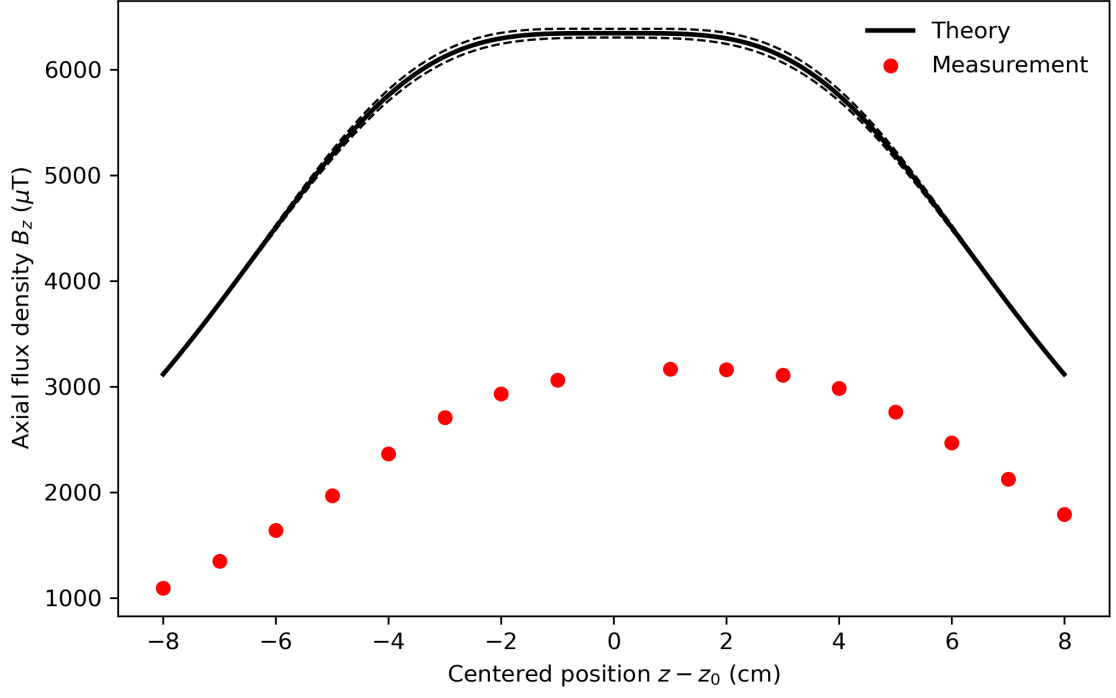


Figure 7: Axial magnetic flux density B_z for the Helmholtz configuration ($b = R$).

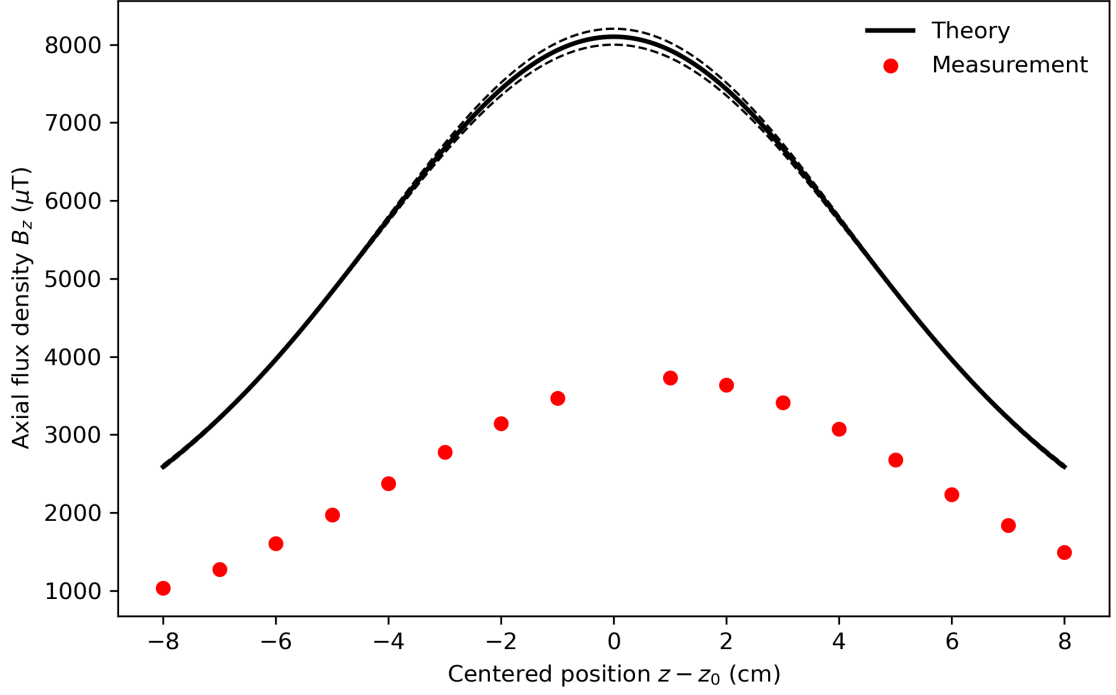


Figure 8: Axial magnetic flux density B_z for a pair of coils with separation $b = R/2$.

The magnetic flux density at the midpoint and the extent of the homogeneous region for the Helmholtz configuration are summarized in Table 3.

Quantity	Value
B_0 (theory, Helmholtz)	6347 μT
B_0 (experiment, Helmholtz)	3113 μT
Homogeneous region (theory)	$[-3.31, 3.31]$ cm
Homogeneous region (experiment)	$[-1.80, 4.11]$ cm

Table 3: Central magnetic flux density and homogeneous region for the Helmholtz configuration ($|B(z) - B_0|/B_0 < 5\%$).

6.5 Uncertainty Analysis

The dominant sources of uncertainty in this task are:

- the finite resolution of the magnetic field sensor (16-bit resolution over a ± 50 mT range),
- the uncertainty of the background field subtraction,
- the current uncertainty of the power supply ($\pm(0.05\% + 3 \text{ mA})$),
- the geometrical uncertainty of the coil radius (relative uncertainty 1.5%).

The experimental uncertainty of the magnetic flux density was obtained by combining the sensor resolution and background subtraction in quadrature, resulting in an uncertainty of approximately $\pm 1.1 \mu\text{T}$. The theoretical uncertainty bands shown in the figures were determined using a Monte Carlo propagation of the uncertainties in the coil radius and current.

6.6 Discussion

For the largest separation $b = 2R$, the magnetic flux density exhibits a pronounced minimum at the midpoint between the coils, as expected from theory. In contrast, for the smallest separation $b = R/2$, a strong maximum occurs at the midpoint, indicating a highly inhomogeneous field distribution.

The Helmholtz configuration ($b = R$) yields the most homogeneous magnetic field in the central region. Both theory and experiment show that the magnetic flux density varies only weakly around the midpoint. The experimentally determined homogeneous region is smaller and slightly asymmetric compared to the theoretical prediction. This can be attributed to systematic effects such as finite coil thickness, small misalignments of the sensor with respect to the coil axis, discretization due to finite position steps, and residual environmental magnetic fields.

Overall, the results clearly demonstrate the special role of the Helmholtz configuration in producing a nearly homogeneous magnetic field and confirm the theoretical predictions within the experimental uncertainties.

7 Discussion

The aim of this experiment was to investigate how different coil geometries influence the spatial distribution and homogeneity of the magnetic flux density generated by steady electric currents. By studying a long solenoid, a single flat circular coil, and a pair of flat circular coils with varying separations, the experiment provides a systematic comparison between distinct magnetic field configurations and their theoretical descriptions.

In Task 1, the magnetic field of a long solenoid was measured along its axis. As predicted by theory, the magnetic flux density inside the solenoid was found to be approximately homogeneous in the central region, while significant deviations occurred near the ends of the coil. These end effects arise from the finite length of the solenoid and lead to a reduction of the magnetic flux density compared to the idealized infinite-solenoid model. The experimentally determined ratio of the magnetic field at the ends to that at the center was of the expected order of magnitude and confirms the validity of the finite-solenoid description. Small discrepancies between experimental data and theory, particularly near the solenoid ends, can be attributed to uncertainties in the coil geometry, imperfect alignment of the magnetic field sensor, and residual background magnetic fields.

The measurements performed in Task 2 highlight the contrast between a long solenoid and a single flat circular coil. In the latter case, the magnetic flux density exhibits a pronounced maximum at the coil center and decreases rapidly with increasing distance along the axis. The experimentally determined half-maximum distance was found to be slightly smaller than the theoretical value. This deviation can be explained by systematic effects such as the finite thickness of the coil, discretization of the position measurements, and the sensitivity of the half-maximum condition to small measurement uncertainties. Nevertheless, the overall shape of the measured field profile agrees well with the theoretical prediction based on the Biot–Savart law.

Task 3 demonstrates most clearly the influence of coil geometry on magnetic field homogeneity. For a large coil separation ($b = 2R$), the magnetic flux density shows a minimum at the midpoint between the coils, whereas for a small separation ($b = R/2$), a pronounced maximum is observed. These observations are in qualitative agreement with the theoretical field profiles obtained from superposition of the individual coil fields. In the Helmholtz configuration ($b = R$), the magnetic flux density varies only weakly around the midpoint, confirming that this arrangement produces the most homogeneous magnetic field among the configurations studied. The experimentally determined homogeneous region was smaller and slightly asymmetric compared to the theoretical prediction, which can be attributed to finite coil dimensions, small positioning errors, and residual systematic effects in the measurement setup.

Overall, the experiment demonstrates good qualitative and quantitative agreement between measured magnetic field distributions and theoretical predictions. Remaining deviations can be consistently explained by experimental uncertainties and non-idealities of the setup. The results underline the importance of coil geometry in the design of magnetic field sources and illustrate why Helmholtz coils are widely used in applications requiring well-defined and homogeneous magnetic fields.

8 Conclusion

In this experiment, the axial magnetic flux density generated by different coil configurations was measured and compared with theoretical predictions derived from magnetostatics. The magnetic field distributions of a long solenoid, a single flat circular coil, and a pair of flat circular coils with varying separations were investigated systematically.

For the long solenoid, an approximately homogeneous magnetic field was observed in the central region, while a clear reduction of the field near the ends confirmed finite-length effects. The experimentally determined ratio of the end field to the central field was found to be consistent with the theoretical expectation. In the case of a single flat circular coil, the strong spatial variation of the magnetic flux density along the axis was measured, and the experimentally determined half-maximum distance agreed qualitatively with the theoretical prediction.

The measurements performed with a pair of flat circular coils demonstrated the decisive influence of coil separation on field homogeneity. Among the three configurations studied, the Helmholtz configuration ($b = R$) produced the most homogeneous magnetic field in the central region, in accordance with theory. The experimentally determined homogeneous region was slightly smaller than the theoretical prediction, which can be attributed to systematic effects such as finite coil dimensions, sensor alignment, and measurement discretization.

Overall, the experimental results confirm the theoretical descriptions of magnetic fields generated by current-carrying coils and highlight the importance of coil geometry in the design of magnetic field sources.

9 References

M. Ziese, E7e Magnetic Fields in Coils , University of Leipzig, December 2025.